

The exponential of a double series

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2026-09-07

Abstract

Module 2 of the coefficient closure for $\eta e^{\beta s \xi}$ (Astra #7(b)). For a coefficient array $a : \mathbb{N}^2 \rightarrow \mathbb{R}$ with $a_{00} = 0$ and $N_\rho(a) < \infty$, the convolution powers a^{*n} satisfy $N_\rho(a^{*n}) \leq N_\rho(a)^n$ and vanish below degree n , so the exponential coefficients $E_k(t) = \sum_{n \leq |k|} t^n / n! (a^{*n})_k$ are finite sums (continuous in t) that agree with the full series. A joint majorant $(n, k) \mapsto |t|^n / n! |(a^{*n})_k| \rho^{|k|}$ with row sums $\leq (|t|X)^n / n!$ gives $N_\rho(E(t)) \leq e^{|t|X}$ (with weighted summability), and grouping the signed family gives $\sum_k E_k(t) u^i v^j = \exp(t \sum_k a_k u^i v^j)$ for $|u|, |v| \leq \rho$. Zero **sorry/axiom**.

1 The things formalised

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def delta :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$  := fun k => if k = (0, 0) then 1 else 0
noncomputable def convPow (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ 
  -- a^{*0} = delta, a^{*(n+1)} = conv a (a^{*n})
theorem wnorm_convPow_le (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (h : 0 < ) (ha : WSummable a)
  (n :  $\mathbb{N}$ ) :
  WSummable (convPow a n) wnorm (convPow a n) wnorm a ^ n
theorem dblSum_convPow (u v :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (h : 0 < ) (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (ha : WSummable a)
  (hu : |u| < ) (hv : |v| < ) (n :  $\mathbb{N}$ ) :
  dblSum (convPow a n) u v = dblSum a u v ^ n
theorem convPow_eq_zero_of_lt (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (ha0 : a (0, 0) = 0) :
  n (k :  $\mathbb{N} \times \mathbb{N}$ ), k.1 + k.2 < n → convPow a n k = 0
noncomputable def expCoeff (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (t :  $\mathbb{R}$ ) (k :  $\mathbb{N} \times \mathbb{N}$ ) :  $\mathbb{R}$  :=
  n Finset.range (k.1 + k.2 + 1), t ^ n / (n.factorial :  $\mathbb{R}$ ) * convPow a n k
theorem expCoeff_continuous (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (k :  $\mathbb{N} \times \mathbb{N}$ ) :
  Continuous fun t => expCoeff a t k
theorem expCoeff_eq_tsum (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (ha0 : a (0, 0) = 0) (t :  $\mathbb{R}$ ) (k :  $\mathbb{N} \times \mathbb{N}$ ) :
  expCoeff a t k = ' n :  $\mathbb{N}$ , t ^ n / (n.factorial :  $\mathbb{R}$ ) * convPow a n k
noncomputable def expMaj (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (t :  $\mathbb{R}$ ) (p :  $\mathbb{N} \times (\mathbb{N} \times \mathbb{N})$ ) :  $\mathbb{R}$  :=
  |t| ^ p.1 / (p.1.factorial :  $\mathbb{R}$ ) * (|convPow a p.1 p.2| * ^ (p.2.1 + p.2.2))
theorem expMaj_summable (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (h : 0 < ) (ha : WSummable a) (t :  $\mathbb{R}$ ) :
  Summable (expMaj a t) ' p, expMaj a t p Real.exp (|t| * wnorm a)
theorem wnorm_expCoeff_le (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (h : 0 < ) (ha : WSummable a)
  (ha0 : a (0, 0) = 0) (t :  $\mathbb{R}$ ) :
  WSummable (expCoeff a t) wnorm (expCoeff a t) Real.exp (|t| * wnorm a)
theorem dblSum_expCoeff (u v :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (h : 0 < ) (a :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (ha : WSummable a)
  (ha0 : a (0, 0) = 0) (hu : |u| < ) (hv : |v| < ) (t :  $\mathbb{R}$ ) :
  dblSum (expCoeff a t) u v = Real.exp (t * dblSum a u v)
```

2 Mathematics

The convolution unit δ has $N_\rho(\delta) = 1$ and evaluation 1, so the closure and multiplicativity of unit 117 give $N_\rho(a^{*n}) \leq N_\rho(a)^n$ and $\text{ev}(a^{*n}) = (\text{ev } a)^n$ by induction. The degree support is the observation that each term $a_b (a^{*n})_{k-b}$ of $(a^{*(n+1)})_k$ vanishes: either $b = 0$ (since $a_{00} = 0$) or $|k - b| < |k| < n + 1$ forces $|k - b| < n$. Hence the finite definition of $E_k(t)$ equals the full series (`tsum_eq_sum`), which is the form needed for the two summation arguments. The majorant $g(n, k)$ is summable on $\mathbb{N} \times \mathbb{N}^2$ because its rows are summable with row sums $|t|^n/n! N_\rho(a^{*n}) \leq (|t|X)^n/n!$, whose sum is $e^{|t|X}$ (`summable_prod_of_nonneg`, `Real.summable_pow_div_factorial`). Then $|E_k(t)|\rho^{|k|} \leq \sum_n g(n, k)$ termwise, and summing over k (after swapping the product order with `Summable.prod_symm` and `Equiv.prodComm`) gives the weighted bound. For the evaluation, the signed family $F(n, k) = t^n/n! (a^{*n})_k u^{k_1} v^{k_2}$ is dominated by g , so `tsum_prod'` regroups it either way: summing k first gives $\sum_n t^n/n! (\text{ev } a)^n = \exp(t \text{ev } a)$, summing n first gives $\sum_k E_k(t) u^{k_1} v^{k_2}$.

3 Paper-facing meaning

With $a = x - x_{00}\delta$ the coefficient array of $\xi - \xi(0)$, the array $E(\beta s)$ is the double series of $e^{\beta s(\xi - \xi(0))}$ on the box, with weighted norm at most $e^{\beta s X_\rho}$. The next unit multiplies by the array of η and the scalar $e^{\beta s x_{00}}$ to produce the s -parametrised coefficient array of the chart amplitude $\eta e^{\beta s \xi}$, with the envelope $Y_\rho e^{\beta s(x_{00} + X_\rho)}$ required by `twoD_cutoff_analytic`: this closes the loop from the paper’s analyticity hypothesis on ξ, η to the analytic $d = 2$ cutoff theorem.

4 Lean notes

- `Real.exp` as a power series: `Real.exp_eq_exp_` followed by `NormedSpace.exp_eq_tsum_div` rewrites $\exp z$ to $\sum' z^n/n!$; the summability is `Real.summable_pow_div_factorial`.
- `(WSummable).mul_left` timed out at `isDefEq` on the folded definition; unfold `WSummable` first and finish with `.congr fun k => rfl`.
- Unreduced projections `(n, k).1` in a summand block `tsum_mul_left`; a `show` of the reduced form first makes the rewrite fire.
- `Nat.abs_cast` in a `simp only` set already removes $|n|$, so a later `abs_of_pos` rewrite fails with “did not find an occurrence”.
- A `rw` chain reports “No goals to be solved” when `rfl` closes the goal mid-chain; drop the trailing steps rather than guarding with `try`.